

MVHR Design Engine

User Manual

HiPer Haus · Passivhaus-Aligned MVHR Sizing for Australian Homes

Introduction

The HiPer Haus MVHR Design Engine is a browser-based tool for sizing and documenting Mechanical Ventilation with Heat Recovery (MVHR) systems in high-performance Australian homes. It follows the Passivhaus Institute (PHI) three-criteria method for calculating governing airflow, selects an appropriate MVHR unit, sizes individual duct runs, and produces a complete commissioning schedule.

All work is saved automatically to your browser as named projects. No internet connection is required once the page is loaded.

Navigation — Top Tabs

The top bar contains eight tabs. Click any tab to switch view. The app stays on the current tab when auto-calculating — you only switch manually.

Tab	Purpose
■ Plan	Upload floor plan, draw building + rooms, place MVHR unit and grilles, auto-route ducts
■ Project	Edit project inputs: floor area, ceiling height, occupancy, duct settings, supply and extract rooms
■ Summary	Design airflow, unit selection, pressure loss and external grille summary
■ Rooms	Room-by-room schedule showing flow rates, duct count, velocity and status
■ Ducts	Individual duct run schedule with ID, length, velocity, pressure loss and status
■ BOM	Bill of materials — all ComfoTube, fittings and terminals required
■ Commission	Commissioning table ready to hand to the installer
■■ Warnings	Design warnings, code issues and informational notes
■ Help	This manual

Project Management

The project bar sits below the navigation tabs and is always visible. Each project stores its own plan drawing, room schedule, and all inputs independently.

Control	Action
Dropdown	Switch between saved projects. Current project auto-saves on switch.
— ■ Rename	Rename the current project.

■ New Project	Save current work, then open a blank project with empty plan and rooms.
■ Copy	Duplicate the current project (plan drawing + all inputs).
■ Save	Explicitly save — a '✓ Saved' badge confirms. Auto-save also runs on every change.

Plan Tool — Step-by-Step Workflow

The Plan tab is the primary input method. Draw directly on your floor plan to set up the entire design. Results auto-calculate as you work.

1 Upload your floor plan

Drag and drop a PDF or image (JPG/PNG) onto the upload prompt, or click to browse. The plan renders to the canvas. For PDFs, the first page is used.

2 Set the scale

Click  Scale in the toolbar. Click two points on the plan that represent a known distance (e.g. a room dimension or north arrow). Enter the real-world distance in metres when prompted. The scale badge updates and all subsequent measurements use this reference.

3 Draw the building outline

Click  Building. Click each corner of the external perimeter. Close the polygon by clicking back on the first point (a snap ring appears) or double-click. The floor area is written directly into the Project tab automatically.

4 Draw supply rooms

Click  Supply Room. Trace the room polygon. On completion a popup appears — select the room type and edit the room name, then click Add Room. The room is added to the Project inputs and an outlet symbol is placed at the centroid.

5 Draw extract rooms

Click  Extract Room. Same process as supply rooms. Extract rooms (kitchens, bathrooms, wet areas) are coloured orange.

6 Mark V-Through zones

Click  V-Through for hallways, stairwells and open-plan corridors. These zones appear on the plan but do not add supply or extract duct runs — they allow air transfer between spaces.

7 Place the MVHR unit

Click  MVHR Unit, then click the plant room or roof void location on the plan. Duct routes are auto-drawn from the unit to every supply and extract room immediately.

8 Place intake and exhaust grilles

Click  Intake and click the external wall or roof location for the fresh-air intake. Click  Exhaust for the exhaust outlet. Dashed duct lines are drawn from the unit to each grille and the external duct lengths update automatically.

9 Auto-route or re-route ducts

Click  Auto Route to regenerate all duct routes. To adjust a route, click any segment to insert a waypoint, then drag the waypoint. Waypoints snap to the MVHR unit and grille positions.

10 Adjust outlets

Drag any outlet symbol (circle or rectangle) within its room to the desired grille position. Click an outlet (without dragging) to toggle between ceiling-mounted and wall/bulkhead-mounted type.

Plan Toolbar Reference

Button	Mode	Interaction
■ Scale	Two-point scale	Click 2 points → enter metres
■ Building	Polygon	Click corners → close to snap ring or double-click
■ Supply Room	Polygon	Click corners → popup to name room
■ Extract Room	Polygon	Click corners → popup to name room
↔ V-Through	Polygon	Click corners → popup to label zone
■ MVHR Unit	Single click	Click to place; auto-routes all ducts
■ Intake	Single click	Click external wall; auto-draws external duct
■ Exhaust	Single click	Click external wall; auto-draws external duct
■ Auto Route	Action	Regenerates all duct routes from unit to rooms
■ Clear	Action	Confirm prompt clears all plan markup

Outlet Symbols

Symbol	Meaning
Blue circle + 4 outward arrows	Ceiling-mounted supply grille
Red circle + 4 inward arrows	Ceiling-mounted extract grille
Blue rectangle + 3 upward arrows	Wall/bulkhead-mounted supply grille
Red rectangle + 3 downward arrows	Wall/bulkhead-mounted extract grille

Project Tab — Manual Input

The Project tab lets you review and edit all design inputs. Values populated by the plan tool can be manually overridden here. Click Run Design Calculation to update results.

Field	Description	Default
Floor Area (m ²)	Total conditioned floor area. Auto-filled from building polygon.	—
Ceiling Height (m)	Average ceiling height across the dwelling.	2.5
Bed Spaces	Number of sleeping spaces (used to estimate occupancy).	—
Bedrooms	Number of bedrooms (informational).	—
Persons	Design occupancy. Defaults to bed spaces if left blank.	Auto
Bends per run	Number of 90° bends assumed per duct run for pressure loss.	0
Supply ext. length (m)	External duct from unit to fresh-air intake grille. Auto-filled from plan.	3
Extract ext. length (m)	External duct from unit to exhaust grille. Auto-filled from plan.	3
External grille type	Individual (separate supply + extract) or Combi (ComfoEdge combined unit)	Individual
Grille / Combi size	Duct diameter of the external grille — selected based on flow.	—

Supply and Extract Rooms

Rooms drawn on the plan auto-populate here. Additional rooms can be added manually using the + Add buttons. Each room card has:

- Room type dropdown — sets the default airflow weighting
- Room name text field — auto-fills from type, edit freely
- Area (m²) — used for context; primary airflow is PHI-calculated
- Duct run length (m) — auto-filled from plan routing; override if needed

Calculation Methodology — PHI Three-Criteria Method

The design airflow is the maximum of three independently calculated values, in accordance with the Passivhaus Institute ventilation standard and PHPP worksheet methodology.

Criterion	Formula	Typical governing scenario
Occupancy-based	Persons × 30 m³/h per person	Small, densely occupied homes
Extract demand	Sum of extract room defaults (kitchen 60, wet areas 40 each)	Homes with multiple wet areas
0.3 ACH minimum	Floor area × ceiling height × 0.3	Large, lightly occupied homes

Airflow Distribution

Individual room flows are distributed proportionally from the governing total using each room type's default weighting. Supply defaults: Single Bedroom 20, Double 30, Master 35, Living 40, Study 20, Dining 25, Rumpus 30. Extract defaults: Kitchen 60, Bathroom/Ensuite/Laundry 40, WC 20, Pantry 15.

Duct Sizing

All ducts are DN90 (90mm diameter) ComfoTube. The number of duct runs per room is calculated on normal flow only: runs = ceil(normal flow / 35 m³/h). At boost (1.3× normal), each DN90 carries up to 45.5 m³/h, keeping terminal velocity below 2 m/s — the PHI limit for acceptable room noise.

Boost Ratio

Boost flow = normal flow × 1.3. The tool displays boost velocity and pressure loss for each duct. Status ratings:

Status	Boost velocity	Meaning
Excellent	< 1.0 m/s	Very low velocity, minimal noise, oversized for application
Normal	1.0–1.5 m/s	Within comfortable operating range
High	1.5–2.0 m/s	Acceptable — approaching PHI limit
Critical	> 2.0 m/s	Exceeds PHI terminal velocity limit — add a duct run

Output Tabs

Summary

Shows the three governing airflow values (normal and boost), the recommended MVHR unit, external pressure loss, and external grille selection. The unit grid highlights the recommended unit in green with a checkmark.

Room Schedule

One row per room. Columns: Room name, type, normal flow (m³/h), boost flow, duct count, terminal velocity, and status. Green = within limits, amber = approaching limit, red = action needed.

Duct Schedule

One row per duct run (S01, S02... for supply; E01, E02... for extract). Columns: Duct ID, room, type, diameter, length, bends, normal flow, boost flow, boost velocity, pressure loss, status. Use this table for ordering ComfoTube lengths and for commissioning balancing.

Bill of Materials (BOM)

Auto-generated list of all ComfoTube DN90 lengths, manifolds, duct terminals (ComfoCase TVA single or twin), silencers, and external grilles. Quantities are calculated from the duct schedule.

Commissioning

A pre-filled commissioning table showing target normal and boost flows for each duct run. Give this to the commissioning engineer. Actual measured values can be recorded in the site copy.

Warnings

Colour-coded flags: Red (critical — action required before installation), Amber (warning — review recommended), Green (ok — within limits), Blue (info — design notes). Common warnings include duct velocity exceeding 2 m/s at boost, external pressure loss outside unit range, and extract demand governing.

Design Tips

Kitchen extract ducts

PHI standard is 60 m³/h for kitchen extract. The tool uses two DN90 runs at 30 m³/h each. Do not combine into one run — boost velocity would exceed 3 m/s.

Multiple rooms of the same type

If you see four 'Living Room' rows, check whether these are truly four separate spaces. If one large open-plan area was drawn multiple times, delete the duplicates and re-draw as one polygon.

Governing criterion

The Summary tab shows which of the three PHI criteria is governing. If extract demand governs, review your wet area count. If 0.3 ACH governs, the home is large and lightly occupied — consider specifying a minimum-flow unit.

External duct length

Longer external ducts add significant pressure loss. Keep intake and exhaust runs under 5m where possible. The plan tool measures the straight-line route — add 10–15% for actual installed bends.

Combi grille vs individual

The ComfoEdge combi grille is ideal where supply and exhaust must exit through the same penetration. Select DN125 for recirculation < 3% or DN160 for < 5%. Do not use combi grilles where the unit is more than 3m from the wall.

Boost mode

Boost (1.3× normal) runs for short periods on demand (cooking, showering). Size for normal flow — the extra capacity at boost is acceptable at velocities up to 2 m/s.

Saving work

The tool auto-saves every 800ms to browser storage. Use  Save explicitly before closing the browser or switching computers. Projects are stored in this browser only — export the PDF as a backup.

Default Airflow Weightings Reference

Room Type	Default (m ³ /h)	Type
Master Bedroom	35	Supply
Double Bedroom	30	Supply
Single Bedroom	20	Supply
Living Room	40	Supply
Rumpus Room	30	Supply

Dining Room	25	Supply
Study / Office	20	Supply
Other (supply)	20	Supply
Kitchen	60	Extract
Bathroom	40	Extract
Ensuite	40	Extract
Laundry	40	Extract
WC	20	Extract
Pantry	15	Extract
Other (extract)	25	Extract

HiPer Haus MVHR Design Engine — Passivhaus-aligned ventilation design for Australian high-performance homes. This tool follows PHI protocol. Always have designs reviewed by a qualified mechanical engineer before installation.